VU AI HUB · LECTURE NOTES

Human-Computer Interaction

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Human-Computer Interaction History (Chapter 1)

> *Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.*

HCI focuses on how humans interact with computers, prioritizing usability, user satisfaction, and the context of the interaction.

- We have computers since the 1940s, but interaction studies since the 1980s.

1.1 Introduction

Human Factors

It's both a science and a field of engineering. It is concerned with human capabilities, limitations, performance, and with the design of systems that are efficient, safe, comfortable, and even enjoyable for the humans who also use them.

- By changing the 'design of systems' to 'design of computer systems' you make the change from human factors to HCI

Time line (last year in exam):

- 1945: Vannevar Bush publishes "As we may think" in the Atlantic Monthly, machines were first for governments, then for everyone.
- 1962: Ivan Sutherland develops Sketchpad (laser mouse).
- 1963: Douglas Engelbart invents the computer mouse.
- 1981: Xerox star brings first computer for commercial use, so normal working people.
- 1982: ACM SIGCHI formed, we need a venue so everyone can get involved with the machines.
- 1983: Card, Moran, and Newell publish The Psychology of HCI.
- 1984: Apple Macintosh launched.
- 2007: 25th anniversary of HCI, the SIGHCI annual conference.

1.2 Vannevar Bush's "as we may think" (1945)

Bush was the US government's Director of the Office of Scientific Research and a scientific advisor to President Franklin D. Roosevelt.

He had several ideas back then that are only now normal to us:

- consequent maze → information overload
- momentarily important item → blog posting or tweets
- memex (navigating the maze) → hyperlinks, bookmarks

1.3 Ivan Sutherland's Sketchpad (1962)

With sketchpads users didn't have to 'write' to the computers by typing, but they could draw with the pen

- There was no user testing for his ideas because he was a student

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1.4 Invention of the mouse (1963)

- Douglas Engelbart invented the mouse

It worked in the same way as the pen, but the pen was annoying to use because you had to lift it to the screen, the mouse was next to the keyboard where your hands already were.

Initially intended for selecting and manipulating text

The user study consisted of comparing the mouse to other input devices like a light pen, joystick, knee-controlled lever and a grafacon (used for tracing curves)

- The mouse error rate was less than half of the others
- Knee-controlled lever was the fastest and the pen after that, but both weren't ideal which caused the mouse to come up on top

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1.5 Xerox Star (1981)

- Commercialization of the mouse only occurred after the Xerox Star launched

It was the first commercially released computer system with a GUI, it had WIMP (windows, icons, menus and a pointing device). It had to be easy to use, so they made the icons resemble normal objects you would use in the office (ex. trashcan).

- Sequential programming: every action occurs in a certain sequence under the system's control (command-line)
- Event-driven programming: accepting the user's action according to the user's wishes in their preferred order (direct manipulation interfaces)

The Star was not a commercial success, because it was more intended for the office environment than for personal use. They would have been perfect, (other options were easy to use but their user interface was terrible and they were very limited) it just didn't reach the right audience, and they were too expensive as well.

1.6 Birth of HCI (1983)

1. *First ACM SIGCHI Conference*

   Special Interest Group on Social and Behavioral Computing (SIGSOC) → Special Interest Group on Computer-Human Interaction (SIGCHI).

   - CHI is a yearly conference for both researchers and practitioners in the HCI field, people submit their research paper and give a presentation if they're accepted

2. *The psychology of human-computer interaction*
    - Book by Card, Moran and Newell

    It has to do with human sensory, cognitive and motor systems
    - perceptual input (the time to perceive a stimulus)
    - cognition (time to decide on an appropriate reaction)
    - motor output (time to react and move cursor)

    They focused on text editing in the book, it demonstrated why and how models are important and how to build them.
3. *Launch of the Apple Macintosh (1984)*

    The ad launched during the Super Bowl and was a great success because of how easy it was to use for personal ends, and it had a mouse with 1 button.

1.7 Growth of HCI and graphical user interfaces (GUI's)

- Microsoft was a late-comer, but now exceeded Mac

1.8 Growth of HCI research

HCI research focused on quality, effectiveness and efficiency of the interface.

Design of menus:

- By using GUI's menus require cognition
- Typing requires recall
    - recognition > recall
- How to organize lots of menu items (depth vs. breadth or function vs. alphabetically)

Human Factors (Chapter 2)

The hardest part are the human factors, humans are more complicated than computers, what works well for one subset of people, won't work at all for another.

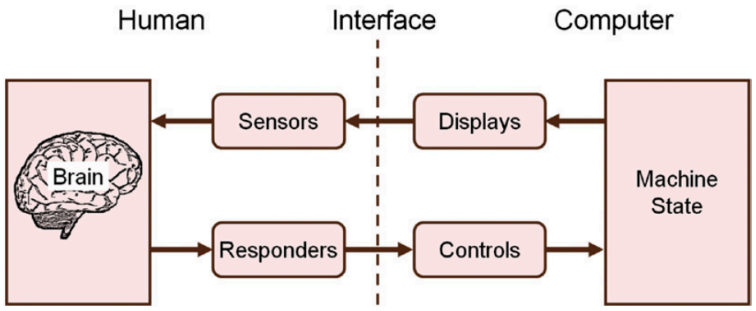
2.1 Time Scale of Human Action

There is a descriptive model of the human by Newell which consists of 4 bands:

- Biological band - experimental and empirical (neural impulses)
- Cognitive band - deliberate acts, operations, and unit tasks (take seconds)
- Rational band - web navigation etc. (take minutes/hours)
- Social band - online dating, social networking (take days)

The most common dependent variable is time in HCI, the time for a user to do a task.

2.2 Human Factors



2.3 Sensors

Vision, hearing, taste, smell and touch

- Reception and conversion into electrical nerve signals of physical phenomena

1. *Vision (Sight)*

   80% of the information we get goes through the sense of light

   - Frequency: property of light leading to the perception of color
     - from violet to red
   - Intensity: the amount of light passing through a given area
     - with luminance^ comes brightness
   - Fixations and saccades: during a fixation, the eyes are stationary, they can be long or short. Changing the point of fixation requires a saccade, which are inherently quick
2. *Hearing (audition)*

   Detection of sound by humans

   - Loudness: equivalent to intensity (how high or low)
   - Pitch: equivalent to frequency, time between peaks in an occurring sound wave
   - Timbre: or richness, unique patterns of harmonies (we can distinguish different instruments)
   - Envelope: the way a note transitions in time, from silent to audible to silent again
3. *Touch (tactition)*

   Part of the somatosensory system

   - Tactile feedback (in HCI) refers to the information provided through the somatosensory system
     - ex. vibration from phones and the F and J keys
4. *Smell and Taste (olfaction and gustation)*
   - Smell cues can aid in recall, because it links to memory
     - but it still performed worse than vision and hearing
5. *Other Senses*

   The word sense appears in many different contexts and encapsulate how humans feel about their interactions with computers (frustrated, satisfaction, confidence, etc.)

2.4 Responders

Control occurs through responders

1. *Limbs*
   - Human control over machines is associated by limbs, particularly the upper body limbs (hands, fingers etc.)

- Handedness (left, right or ambidextrous) is relevant in situations involving touch- or pressure-sensing displays

2. *Voice*
   - Speech must be recognized by algorithms implemented in software
   - Good for accessibility
3. *Eyes*
   - Both a sensor and responder
   - Eyes can be used as a mouse by eye-tracking apparatus

2.5 The Brain

The brain connects the sensors and responders

1. *Perception*

   This is the first stage when sensory signals are received as an input
   - Psychophysics - examines the relationship between human perception and physical phenomena
   - Ambiguity shows our perceptual ability to go beyond the given information
     - illusion is the deception of common sense and shows that everyone can perceive things differently
2. *Cognition*

   The human process of conscious intellectual activity (such as thinking, reasoning and deciding)
   - Cognitive operations are hard to research, because you can't pinpoint when the sensory stimulus and motor response happens
3. *Memory*

   The human ability to store, retain and recall information
   - Human memory and computer memory are surprisingly similar organized, with different sections for different kinds of information
   - Chunking is done by humans to make lots of information easier to remember (063818700 → 06 38 18 78 00)

2.6 Language

Mental faculty that allow humans to communicate. In HCI, the interest in language is primarily in systems of writing and in the technology that enables communication in a written form

1. *Redundancy in language*

   Native speakers of a language have an immense understanding to the point where they can fill in missing words/letters
2. *Entropy in language*

   The uncertainty about forthcoming letters/words/phrases/ideas/concepts/etc.
   - Errors are more common at the beginning of words and become less when they progress

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2.7 Human Performance

When humans use their sensors, brain and responders, the performance arises

- Humans bring diversity and variability which then brings imprecision and uncertainty in HCI (everyone is different)

1. *Reaction Time*

Delay between the occurrence of a single fixed stimulus and the initiation of a response assigned to it

- Physical matching: if the words have the same amount of letters
- Name matching: if the words are the exact same, ignoring different fonts and styles
- Class matching: if the characters are both letters or both digits
- Choice reaction: chapter 7

2. *Visual Search*

   Scanning a collection of items, searching for a desired item

   - Takes longer when there are more items to choose from

3. *Skilled Behavior*

   Property of human behavior whereby human performance necessarily improves through practice

   - Human-machine interaction requires both sensory-, motor- and mental skills

4. *Attention*

   Property of human behavior where if someone attends to one thing they can't attend to another

   - while this applies to typing for example, it doesn't to walking
   - Divided attention: concentrating on more than one task at a time
   - Selected attention: or focused attention, is concentrating on one task and excluding others
   - Selective attention: only reacting on your own name for example

5. *Human Error*

   Discrete event in a task, or trial, where the outcome is incorrect, having deviated from the correct and desired outcome

   - The how and why questions don't only apply to humans but they're also about the device and the interaction

Prototypes (Lecture 2)

> "A concrete representation or part of all of an interactive system"

We prototype to:

1. minimize the time for early design iterations
2. make errors quickly
3. enables to explore more design options
4. learn the underlying skillset

sketches → wireframe (an idea of how you want to show things on the display but without the images etc.) → full mockup → software design

- Low fidelity prototype: when you don't have a goal, but just trying things out (starting point of prototypes)
  - Cheap, easy and quick to implement
  - Users are keen to criticize
  - It has no real functionality
  - Difficult to identify errors
- High fidelity prototype: almost fully finished prototype, missing some hidden features
  - Looks and feels like the final product
  - Only certain functions work
  - There are several invisible issues

- Horizontal prototyping: when you have an idea of what you want to do, but don't know the small features work
- Vertical prototyping: trying to prototype a specific feature
  - Often the selling point of a product to distinguish yourself from others on the market
- Non-functional prototypes
- Functional prototypes
- Evolutionary prototypes: start with a prototype and keep on making it better
- Throw-away prototypes: you build it once, but when actually building you scratch everything and start anew

Interaction Elements (Chapter 3)

Interaction occurs whenever a human uses technology to perform a task. These tasks can be specific and goal-oriented (like programming or a thermostat) or open-ended (like browsing social media).

- Cognitive band: researchers focus on actions that last between 100ms and 10 seconds, because they're easier to study

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3.1 Hard controls and soft controls

- Hard controls: physical, single-purpose devices (joysticks, switches, etc.)
- Soft controls: software-rendered elements (buttons on a toolbar), flexible and easy to change their interface
  - They allow for high density, over 20 controls can exist on a small surface. However, this leads to user errors because all controls are close to each other.
    - a single action (ex. click) can completely change the entire system interface, changing what the user sees and how to interact
- Blurred lines: in software things can be both a control and a display, like a scrollbar
- Space multiplexing: every function has its own dedicated physical button in a specific location
- Time multiplexing: a single physical device is used to control different parameters at different times

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3.2 Control-display relationships

Sometimes called mappings, because it shows how a controller property maps to a display property (moving your mouse, movement on screen with cursor)

1. *Spatial relationships*
   - Congruent mapping: controller input = display output
   - Transformed mapping: motion is not identical (moving a mouse forward results in the cursor going up)
   - The 6 Degrees of Freedom: to fully place and orient an object in 3D space, 6 parameters are needed:
     - Position (translation): x, y, z - axes

       → tells the computer where the object is or where it is moving
     - Orientation (rotation): pitch (θx), roll (θz), and yaw (θy)

       → shows where an object is
   - Camera movements:
     - Panning: rotating the camera left-right and up-down
     - Zooming: changing magnification of the view

        - Positioning: change in the camera position
2. *CD gain and transfer function*
    - CD gain: the relationship between how far you move a controller (like a mouse) and how far the object moves on the screen
        - CD gain = 1 when mouse move = 3 cm and cursor move = 3 cm
        - CD gain = 2 when mouse move = 3 cm and cursor move = 6 cm
    - Transfer function:
        - Gross positioning: gain increases to help cover large distances quickly
        - Fine positioning: gain decreases to give you pixel-perfect accuracy

    Optimizing both speed and accuracy is hard because of the trade-off: what improves speed tends to worsen accuracy and vice versa

        - Fitt's throughput: complex metric to combine the two into a single score to see if the CD gain gets better
3. *Latency*

    Time delay between a user's physical action and the system's corresponding response on a display

    - Closed-loop: we issue a command and wait for a response, if that response is delayed, the loop is broken, and our performance drops
    - Latency (or lag) is usually unnoticeable with tasks like typing, but it becomes extreme in remote manipulation (something on Mars)

    User frustration can be reduced by showing a 'processing' notification

    Sources of latency:

        - Input devices: controllers are sampled at fixed rates
        - Software overhead: network configurations, and the way the application processes add to the wait time
        - Graphics rendering: to achieve real-time animation takes time
        - Double buffering: a computer technique where it waits until a frame is fully drawn before showing it, this can add lag

    If there is too much lag in VR it breaks the illusion, researches use external high-speed cameras and tools to see the gap between the physical move and the screen update

        - low latency is more important that high-quality graphics
        - high lag is a primary cause of motion sickness
4. *Property sensed and order of control*

    Input devices act as transducers, converting physical movement or pressure into digital coordinates, 3 most common properties:

        - Position: absolute coordinate (ex. x or y) on a surface, typical for touchscreens
        - Displacement: relative change in movement, a mouse uses displacement because it doesn't know where it is on a desk, only how far and fast it's moving
        - Force: amount of pressure applied to a non-moving part

    Order of control: how the senses property is mapped to the behavior of the object on the screen

        - Position control (Zero-order): the input determines the location of the screen object
        - Velocity control (First-order): the input determines the speed and direction of the screen object

    Order of control is often associated with joysticks, which can be:

        - Isotonic (displacement joysticks) where the stick moves freely
            - low resistance and you know where the stick is, because you can feel its position

        - best paired with position control
    - Isometric (force joysticks) where the stick is rigid
        - you need force/pressure (ex. pointing stick in some keyboards)
        - best paired with velocity control

3.3 Natural versus learned relationships

A relationship is natural if the physical movement of a control exactly matches the physical movement or expected outcome on the display

- Moving a physical slider up and the digital bar on the screen moves up

A relationship is learned whenever these is a spatial transformation between the control motion and the display response, even if it feels simple and intuitive

- Rotating a round dial to make a linear bar move up

- Population stereotype (or cultural standards): A learned relationship that is operated and considered correct by a majority of people within a specific cultural, ethnic, or geographic group.

*Physical analogy: a method of understanding a complex system by comparing it to a more familiar physical system that shares similar structural or mathematical properties

- When a task lacks a physical analogy, population stereotypes arise based on regional consistency. Ex. flipping a light switch up means on in the US but it means off in the UK
- Warrick's principle: a display indicator should move in the same direction as the nearest point on the control
    - the pointer on a knob should move in the same direction as the vertical meter next to it for example
- When a control and its response are physically misaligned (ex. placing an Up button below the Down button) it can be learned but will result in slower reaction times and more mistakes
- HCI relies on metaphors to help users to understand a system, because there is no physical analogy for saving a file for example

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3.4 Mental models and metaphor

- Mental models: the internal understanding a user has about how a system works, a successful model aligns with the user's mental model
    - like the Desktop Metaphor: trashcans, folders, etc.
- Implementation models: opposite of mental models and should be avoided, the system forces the user to interact with it based on how the software's underlying code works
- Icons like magnifying glass for zoom work as mental triggers
    - Tooltips (small pop-up text when you hover over the icon) can make their purpose even clearer
- Directional metaphors: Using North, South, East and West for directions or a clock (ex. playing a sound in your right ear to indicate you should turn right in GPS)

3.5 Modes

A functioning arrangement or condition in a system where the exact same user input produces a different outcome depending on the active state (ex. pressing A gives a normally, but A in Shift mode)

- They exist because systems almost always have fewer physical controls than functions, modeless designs are practically impossible (even a car needs Neutral, Drive, and Reverse modes)
- Mode errors occur when a user forgets or doesn't realize which mode the system is currently in

    - to precent these errors, systems must provide clear, obvious, and continuous feedback to the user about the mode they're in (ex. you're in Caps Lock)
- Making a mouse able to rotate on the screen when twisting the mouse (detecting the twisting with sensors or cameras) leads to a confusing mode and lots of trial-and-error

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3.6 More about Degrees of Freedom

Designing input devices isn't just about how many DOF they have, but how they relate to the task at hand. They can be integrated or separated

- Integrated (combined) DOF: your hand/fingers control multiple dimensions at once in one fluid motion, easy to move diagonally, but hard to move straight in any direction
    - Joysticks on game controllers
- Separated DOF: each dimension gets its own dedicated physical control, easy to move straight in any direction, harder diagonally
    - Keyboard arrow keys (impossible to move diagonally)

You can add separate DOF's like the scroll wheel on the mouse, it separates 1 DOF from the rest

- A rockin' mouse: which senses standard movements, but also side-to-side (Oz) and front-to-back (Ox) tilting
- Trackball: senses rotation easily
- Under-Display touch: placing touch sensors on the back of a mobile device, basically a full laptop trackpad on the back of your phone to you didn't have to touch the screen

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3.7 Mobile Context

- Indirect input: the system response is spatially displaced from the input device (a cursor and a mouse)
- Direct input: used by touchscreens, system response is exactly located at the point of a physical touch
    - precise pixel-level on a touchscreen is very hard, fingers are big

Touchscreen selection methods:

- Land-on: selection occurs the instant the finger touches a target
    - tapping an app icon on your home screen
- First-contact: selection occurs when a sliding finger first enters a target zone
    - pattern lock on Android
- Take-off selection (offset cursor): a popup appears slightly above the finger, you can drag the finger until the offset cursor is on the target, then lifting the finger to confirm
    - When typing the letter that your finger is on gets shown in a bubble above your finger, so you know what you're typing
- Multi-Touch: sensing multiple points of contact, like zooming in or out with 2 fingers

Modern smart devices are packed with components that act as responders (cameras, light sensors, etc.). Allowing the device to sense its own tilt can make a character move in a game

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3.8 Interaction Errors

Severe errors usually get fixed quickly. However, small errors often slip past programmers and linger in systems, repeatedly frustrating users, here are 4 examples of interaction errors (from severe to minor):

1. Clicking on Discard changes instead of Save changes
2. Not knowing if Caps Lock is on
3. Highlighting text and dragging the mouse is often poorly tuned

4. Not knowing if the cursor will jump to the next box when filling out a form, because there is no consistent standard

User experiences are built on tiny details and micro-strategies, when programmers ignore there to focus on larger things, the user experience suffers

- Often called user errors, but actually design errors for not accounting to natural human behavior and capabilities

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Additional Lecture Notes

Classification of Pointing Devices

Direct vs Indirect

- Integration with the visual representation
  - touch screen is direct
  - mouse is indirect

Discreet vs Continuous

- one single unit vs a sequence of information interpreted as one input action
  - push button is discreet
  - mouse is continuous (it's fast enough to be continuous)

Absolute vs Relative

- movement/position used as input
  - touch screen is absolute
  - mouse is relative

Scientific Foundations (Chapter 4)

4.1 What is Research?

Research is a careful or diligent search, collecting information about a particular subject or investigation/experimentation aimed at the discovery and interpretation of facts and revision of accepted theories or laws in light of new facts.

- Empirical research is a broad term that encompasses both experimental methodologies and non-experimental ones (like building/testing models of interaction)

1. *Research must be published*
   - Publication needs to be done, it needs to be published in archives, peer-reviewed, journals or conference proceedings
   - Patenting an invention is also a form of publication, as it discloses the invention to the public
2. *Citations, references, impact*
   - Citations connect research papers together, showing how new research builds on or refines older research
   - The quality and significance of a paper are generally measured objectively by its citation count
     - = how many other papers cite this one
3. *Research must be reproducible*
   - A standardized methodology (= a set of instructions about your research so others can recreate it) is crucial because it ensures the researcher provides enough detail for others to reproduce or verify, without this ability the research is useless

4. *Research versus engineering versus design*
    - Engineering/Design focuses on building complete products, balancing form (design) and function (engineering)
    - Research narrowly focuses on conceiving, prototyping, and testing small, incremental ideas that engineers later use
        - It precedes engineering and design, because the idea for things were there before the product

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4.2 What is empirical research?

Research should be guided by direct observations and experiences about phenomena, without prejudice to, or even consideration of, existing theories. Another definition says that empirical means capable of being verified or disproved by observation or experiment.

4.3 Research methods

There are three common empirical approaches for research in HCI:

1. *Observational method*
    - Include things like: interviews, field investigations, case studies
    - Generally focused on the why or how of an interaction (discovering thoughts, feelings, attitudes and emotions) rather than quantifiable human performance
    - They're studied in natural real-world settings, which means it achieves high relevance but sacrifices precision
2. *Experimental method*
    - aka scientific methos, knowledge is acquired through controlled experiments conducted in laboratory settings
    - This setting eliminates real-world chaos making it high in precision, but it diminishes relevance

    It requires at least 2 variables:

    - Manipulated variable (independent variable/factor): it is deliberately changed or controlled by a researcher to test its effect on a dependent variable
    - Response variable (dependent variable): the main factor measured, observed, or tested

    A 'usability evaluation' usually looks at a single interface for weaknesses (no manipulated variable = no experiment). A 'user study' is a controlled experiment testing and comparing multiple factors.

        - The experimental method is the only one that allows researchers to draw powerful cause-and-effect conclusions
3. *Correlational method*
    - Looking for relationships between existing variables
    - It requires quantification (measuring magnitudes like income) or categorization (nominal scales like gender). Data is gathered via surveys, questionnaires, interviews or observation
    - There's a balance between relevance and precision
    - It can only prove circumstantial relationships exist, no cause-and-effect conclusions

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4.4 Observe and measure

Observation is the foundation of empirical research

1. *Observation*
    - Human observers manually write down events or use a stopwatch to time tasks, often inaccurate but best suited in public spaces

    - Computers have custom software that easily and accurately logs timestamps, keystrokes and mouse movements
        - but they can't observe everything, so they need human observers or secondary equipment (like eye trackers)
2. *Measurement scales*

   There are four scales scales of measurement, from least sophisticated to must: nominal, ordinal, interval, and ratio
3. *Nominal*
    - Assigning a code to an attribute or mutually exclusive category (ex. gender: M or F)
    - The only relationship it holds is equivalence, mathematical equations are completely meaningless
        - Often used with frequencies or counts to compare categories
4. *Ordinal data*
    - Provides an order or ranking to an attribute
    - The interval/distance between successive points is not intrinsically equal or known (ex. 1st is preferred over 2nd, but you don't know to what extent)
    - Allows for 'greater than' or 'less than' comparisons, but calculations are invalid
5. *Interval data*
    - Data where the distances between adjacent values are exactly equal, but there is no absolute zero (ex. degrees Celsius)
    - They're not meaningful (ex. 20° is not 2x hotter than 10°)
    - Commonly used in questionnaires via Likert Scales (ex. a 1-5 scale from 'Strongly Disagree' to 'Strongly Agree'), because the gradations are treated as equal you can do calculations
6. *Ratio data*
    - It has an absolute zero and equal intervals, meaning all mathematical calculations are valid
    - The most common ratio measurement in HCI is time (task completion time). It also includes physical measurements (cursor velocity, finger force) and demographic variables (age, experience, etc.)
    - Counting occurrences yields ratio data. However, raw counts are often unhelpful for comparison, they must be normalized (a rate)
        - typing 10 words in 30 seconds → 20 words per minute

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4.5 Research questions

Experimental research answers questions about the relationship between a manipulated interface property and a measured human response (task performance)

- Computers are predictable, humans are very unpredictable, even when repeating the same task
- Avoid vague terms like 'better than' or 'any good', because they're untestable since they cannot be objectively measured

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4.6 Internal validity and external validity

When formulating research questions, there is a constant tension between how accurate your results are and how broadly they apply in the real world

- Internal validity: the extent to which an observed effect is confidently caused by the test conditions, rather than by outside factors (like distractions, time of day or human unpredictability)
    - High interval validity means you are highly accurate, you can confidently say your manipulated variable caused the changes
    - Narrow, testable questions yield high internal validity

- External validity: the extent to which your experimental results are generalizable to other people and other situations
  - High external validity means your findings are broadly applicable outside the lab
  - Broad, untestable questions have high external validity because they encompass the whole user experience, but they're impossible to measure accurately

In practice, improving one validity usually damages the other

→ Researchers handle this by posing multiple narrow, testable questions

- Ecological validity refers to the methodology (setting up a realistic environment and using real-world tasks). External validity refers to the outcome

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4.7 Comparative evaluations

Designing and testing just one new interface is weak research. True experimental research requires an independent variable with at least two conditions to be able to make comparisons.

- Always compare your new design against an established standard, this provides two major benefits:
  1. Methodology check: Since the baseline has been heavily studies, you know how it should perform. If your results match past research, you know your testing setup is solid
  2. Cross-study Comparison: A universal baseline acts as an anchor, allowing researchers to compare new designs across completely different studies that share the same baseline

Comparative evaluations actually change user behavior. Research proves that when users test multiple interfaces side-by-side, they find more flaws, are more critical, and give much more realistic ratings comparing to just testing one interface in isolation.

4.8 Relationships: circumstantial and causal

- Causal relationships: Found exclusively through controlled experiments with random assignment (cause-and-effect)
- Circumstantial relationships (correlational): It shows two variables are linked, but cannot prove one caused the other due to uncontrolled variables

You can never claim cause-and-effect for naturally occurring human traits (like gender, personality, etc.). Because you cannot randomly assign someone to be 'male', you can never separate this from the confounding variables inherently attached to it.

Additional Lecture Notes:

Usability: effectiveness, efficiency, learnability, memorability, error prevention

User experience: satisfaction, enjoyment, pleasure, fun

The dependent variable is always the human

Designing HCI Experiments (Chapter 5)

Every observation or measurement you take in an experiment is a combination of two components:

- Signal - this is your variable of interest (ex. testing a new input device), it is the underlying truth or trend you are actually trying to measure
- Noise - these are all the random, unwanted influences that cause messy variability in your data
  - Environmental noise: Room temperature, glare on the screen, background noise, or a wobbly chair

- Human noise: A participant having a bad day, feeling tired, or just natural variations in how different people behave

You want to carefully construct your methodology in a way that maximizes the signal while minimizing the noise.

5.1 What Methodology?

The term 'method' or 'methodology' refers to the comprehensive, step-by-step plan of exactly how an experiment is designed and carried out. Establishing a methodology means making deliberate choices in advance about:

- The people involved (participants)
- The hardware and software used (materials or apparatus)
- The tasks they will perform, and the exact order of those tasks
- The procedure for briefing and preparing the participants before they start
- The variables (what you are manipulating vs. what you are measuring)
- The exact data that will be collected and how it will be analyzed

---

5.2 Ethics approval

HCI involves real human beings, so securing ethics approval is a mandatory step that must happen before an experiment is fully designed or conducted

- The core purpose is to ensure researchers respect the safety, welfare, and dignity of their participants. The review committee strictly ensures that participants are fully informed of their rights before starting, these include:
  - Knowing the exact nature, goals, and hypotheses of the research
  - Knowing the methodology (what they will specifically be asked to do)
  - Knowing any potential risks or benefits
  - The right to withdraw
  - The right to have their data kept completely anonymous and confidential

---

5.3 Experiment Design

The process of bringing together all the necessary pieces; variables, tasks, procedures, and participants, to test hypotheses about a user interface or interaction technique.

Thinking about experimental variables forces us to craft narrow and testable questions, the two most important ones are dependent and independent variables.

5.4 Independent variables

Also called a factor, is a specific circumstance or characteristic that the researcher manipulates or controls to see if it causes a change in human performance.

- It must always have at least two levels (the specific variations you're testing). Ex. if the independent variable is 'Device', the levels might be 'mouse', 'trackball', and 'stylus'.

  *It's called independent, because it's completely independent of the user's behavior, a participant can't do anything to change or influence the fact that they're testing a mouse right now.

Types of independent variables:

- System attributes - things easily manipulated in the software/hardware (ex. display size, menu type)

- Human characteristics - naturally occurring traits of the user (ex. age, gender, handedness), you can't actually manipulate these but you can group them based on these things
- Environmental circumstances - external factors like background noise or room lightning

Always name the independent variable and its levels clearly and once you picked a name for them, do not change them later in the paper, stay consistent.

With every additional variable, the number of statistical effects multiplies exponentially, which makes it harder to find meaningful interpretations for all the effects.

5.5 Dependent variables

A measured human behavior, called dependent because the measurement depends entirely on what the participant does during the task. In HCI the most common dependent variables focus on speed (aka task completion time) and accuracy (aka error rate).

> *Any observable, measurable aspect of human behavior can be a valid dependent variable, provided it helps expose differences between your test conditions.

Always state the name of the variable separately from its unit of measurement (ex. variable = text entry speed, unit = words per minute)

The most efficient method is to program your experimental software to automatically log timestamps, key presses, and events directly into a data file. The data should be organized and stored in a way that makes sure that follow-up analyses are possible.

5.6 Other variables

Next to the dependent and independent variables, every experiment has three other types of variables that researchers must manage:

1. *Control variables*
    - circumstances or factors that might influence your measurements but are not the focus of your research
        - you control them by locking them in at a fixed, nominal setting so they don't interfere with the experiment
        - ex. room temperature, lighting, background noise, mouse cursor speed, etc.
    - Using control variables increases internal validity (because locking down variables proves your independent variable caused the effect), but compromises external validity (because you limit your results to a very specific, artificial situation or type of person)
2. *Random variables*
    - factors that you deliberately allow to vary naturally instead of trying trying to control them, typically human traits
        - ex. participants height, hand size, IQ, etc.
    - Using random variables increases external validity (because testing a wide variety of people makes the results more generalizable to the real world), but compromises internal validity (because you introduce extra messy variability into your data)
3. *Confounding variables*
    - any circumstance or condition than changes systematically alongside your independent variable
        - these are the biggest threat to experimental research, because you can't prove that your independent variable caused the change in performance
        - ex. testing Interface A first, then Interface B. The user's natural improvement (practice) systematically favors Interface B

5.7 Task and procedure

When a participant tests an interface, they must be given a specific task to perform so you can measure their response. The goal of this task is to elicit behavior that exposes the differences between your test conditions.

There are two objectives of a good task:

- Represent: The task should mimic real-world or expected usage, this improves external validity
  - making a task too representative leads to a lot of noise, that are likely to compromise the ability of the task to discriminate among the test conditions and it also decreases the internal validity
- Discriminate: The task must actively highlight the specific differences between the interfaces being tested

The task should feel natural but focus strictly on the points of differentiation, if something works the same way in both interfaces you are comparing, you should exclude it from the task to reduce unnecessary noise.

The procedure is the overarching timeline of the experiment, it includes the task itself, but also the instructions given, demonstrations, practice rounds, and any pre- or post-experiment questionnaires.

5.8 Participants

To ensure your results apply to the real world, you must test people who actually belong to your target population, and you must test enough of them.

You have to look at similar published research and use the same number of participants to test as they did

- Using too many: you might achieve statistical significance for tiny differences that don't actually matter in the real world
- Using too few: you might fail to statistically prove that a genuinely better interface actually works
- Exception: if you are just testing a prototype to find bugs (usability testing), ~5 participants is usually enough to catch 80% of the problems

Truly random sampling is hard, so researchers use convenience sampling (recruiting whoever is nearby, like local students). This is standard practice, even though it technically lowers external validity.

Use the word 'participants' when describing the experiment, and 'users' when making broad conclusions about the real world.

5.9 Questionnaire design

HCI questionnaires are usually very brief and serve two specific goals:

1. Gathering basic demographic information and prior technology experience
2. Soliciting the participants' subjective opinions about the interfaces they just tested

Types of questions:

- Closed-ended: forces the user to pick from a constrained set of options, highly efficient because they simplify your follow-up analysis by making it easy to go through the responses
- Open-ended (ratio-scale): asking the user to provide an exact number (like asking for their age), this yields high-quality, ratio-scale data because it allows you to calculate the Mean and Standard Deviation
- Ordinal ranges: Asking a user to select a range (ex. Age: 20-29), this provides inherently lower-quality data than asking for the exact number

To measure a user's subjective experience, researchers typically use standard 1-to-7 Likert scales

  - NASA-TLX: used to assess perceived workload (1 is the best outcome, while 7 is the worst)
  - ISO 9241-9: used to assess user comfort (1 is the worst outcome, while 7 is the best)
- Keep the scales consistent, either keep 7 as the best option or 1, do not mix them up

5.10 Within-subjects and between-subjects

- Within-subjects: every participant tests all condition
  - must use this when studying practice or learning over time
- Between-subjects: separate groups of participants test only one condition each
  - must use this for human traits (ex. gender or handedness)

Between-subjects requires many more people, while within-subjects requires more time per person

5.11 Order effects, counterbalancing, and latin squares

Order effects: in within-subjects designs, the order of testing creates interference, performing better in the second task because of practice or worse because of fatigue.

Counterbalancing: by dividing participants into equal groups and giving them the tests in different orders you fix this problem.

Latin squares: standard mathematical grid for counterbalancing, where each test condition appears exactly once in every row and column

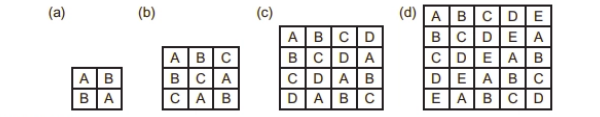


FIGURE 5.7

Latin squares: (a) 2 × 2. (b) 3 × 3. (c) 4 × 4. (d) 5 × 5.

Balanced latin squares: ensures every sequence is perfectly equal, in normal latin squares Condition B might follow A three times, but A only follows B once

- you can't perfectly balance an odd number of conditions, you must use every possible sequence combination (ex. 3! = 6)

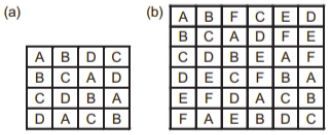


FIGURE 5.8

Balanced Latin squares where each condition precedes and follows other conditions an equal number of times: (a) 4 × 4. (b) 6 × 6.

5.12 Group effects and asymmetric skill transfer

Group effects: counterbalancing only works if the learning effect is identical regardless of test order, if the two groups end up with totally different overall averages, counterbalancing failed

Asymmetric skill transfer: this is the cause for these fails, it happens when one test condition acts as a perfect 'training round' for the other

- Because users can't 'unlearn' a skill, if one interface heavily trains the users for the other, you must abandon within-subjects and use a between-subjects design instead

5.13 Longitudinal studies

Instead of trying to eliminate learning effects, a longitudinal study actively measures them by testing participants over a prolonged period of time

- In these studies, the amount of practice becomes the independent variable

- They're used to prove that a completely new, unfamiliar interface is ultimately better than an established current practice
  - Because users already have years of experience with the old system, their initial performance on the new system will naturally be worse. However, if the new design is mathematically better, it will eventually beat the old system's performance
    - Only after the crossover point, where the user learned to work with the new interface, will it provide benefits

---

5.14 Running the experiment

- Before officially starting your study, run one more pilot test with 1-2 people. This isn't to test the software, but to test your protocol, ensuring your briefing is smooth and the session comfortably fits within the scheduled time
- A typical session follows a strict order: Greet → Sign consent forms → Demographic questionnaire → Reveal the apparatus → Explain/demonstrate the task → Do practice trials → Begin actual testing
- You must deliver the exact same instructions to every single participant
  - if a user asks for clarification, be extremely careful not to accidentally add new information that makes them behave differently than the other participants
- Remain neutral as the researcher

Hypothesis Testing (Chapter 6)

Statistics can be a controversial and highly theoretical topic. In HCI research, the focus is generally practical, understanding what the test means, how to run it, and how to explain the results, rather than memorizing the complex math behind it.

There are two different kinds of statistical tests you can use, the choice entirely depends on the scale of measurement of your data

1. Parametric Tests (for Interval & Ratio data)
   - Tests that assume your data follows a specific probability distribution (like a normal/bell-curve distribution)
   - Used for high-level, quantifiable human performance data (ex. task completion time, speed, error rates, etc.)
   - The Analysis of Variance (ANOVA) is the most widely used parametric test in HCI
   - They possess higher statistical power, meaning they have a greater ability to draw correct, confident conclusions, provided the data meets the strict assumptions
2. Non-Parametric Tests (for Nominal & Ordinal data)
   - 'Distribution-free' tests that make zero assumptions about the underlying distribution of your data
   - Used for categorical data, questionnaire responses, and subjective rating scores
   - Chi-square, used for nominal data
     - Mann-Whitney U, Wilcoxon Signed-Rank, Kruskal-Wallis & Friedman, used for ordinal data
   - They are highly flexible and can technically be applied to any measurement scale, though they lack the raw statistical power of parametric tests

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6.1 Analysis of Variance (ANOVA)

The primary statistical procedure used in factorial experiments. It determines if an independent variable (the test conditions) had a statistically significant impact on a dependent variable (the measured human response)

It compares two things: the variance (difference) *between* the test conditions vs the variance caused by everything else (the participants, random noise, etc.). If the variance caused by test conditions is significantly larger, we can conclude that they caused the difference in performance.

1. *Why analyze the variance?*
    - ANOVA begins by assuming there is no difference between the test conditions (ex. Method A and Method B take the exact same amount of time)
    - The goal is to prove the null hypothesis is highly unlikely, thereby proving a difference actually exists

      *You can have two test conditions with different average scores, but if the participant's individual scores are wildly inconsistent (huge variance/error bars), the difference in the average is meaningless, it could just be chance. ANOVA analyzes that variance to see if the difference is real.
    - Key statistics:
        - P-value: the probability that the observed differences were just a chance outcome. In HCI, the threshold for 'statistical significance' is $p < .05$ (less than a 5% chance the null hypothesis is true). If $p > .05$ it is not significant.
        - F-statistic: the ratio of sample variances, it is reported with degrees of freedom
    - ANOVA tells you *if* a difference exists. Effect size tells you *how big* that difference was in plain language (ex. Method A was 20.1% faster than Method B)
2. *More than two test conditions*
    - ANOVA can handle experiments with more than 2 test conditions at once (ex. comparing 4 different interface designs)
        - However, if the result ends up to be statistically significant, it only tells you that at least one condition is significantly different from at least one other condition. It doesn't tell you which ones are different.
3. *Post hoc comparisons*
    - When testing 3+ conditions, if the ANOVA is significant, you must run a post hoc comparisons test (Scheffé test) to compare each condition against the others (A vs B, A vs C, B vs C) to pinpoint exactly where the significant differences lie.
4. *Between-subjects designs*
    - ANOVA handles both within-subjects (all participants test all conditions) and between-subjects (different groups test different conditions) designs.
        - the math is slightly different because between-subjects data has different sources of variance
5. *Two-way analysis of variance*
    - If an experiment has two independent variables, you use a two-way ANOVA, it tests for three things; if the first variable mattered, if the second variable mattered and did a specific combination matter (interaction effect).
6. *ANOVA tool*
    - To help us run statistical tests, check the book
7. *Counterbalancing and testing for a group effect*
    - If you used counterbalancing (Group 1 does A then B; Group 2 does B then A) to prevent learning effects, you can use ANOVA to double-check if it actually worked
        - you treat 'Group' as a between-subjects factor and run an ANOVA, if the main effect of 'Group' is not significant your counterbalancing successfully canceled out the learning effect

6.2 Chi-square test

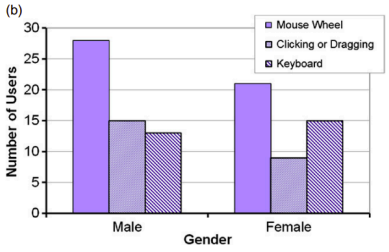
A non-parametric statistical test used to investigate relationships between categorical (nominal-scale) variables. You use this when your data consists of counts of frequencies of people, methods, or systems that fall into specific categories.

- Data is summarized in a contingency table:

(a)

Observed Number of Users				
Gender	Scrolling Method			Total
	MW	CD	KB	
Male	28	15	13	56
Female	21	9	15	45
Total	49	24	28	101

(b)



- The test compares the observed values (the actual counts you recorded) against the expected values (the hypothetical counts that should exist if there were absolutely no difference or relationship between the categories)
- Like ANOVA, the starting assumption is that there is no difference between the categories
- Determining the significance:
  - You must calculate the Degrees of Freedom (df): (rows - 1) x (columns - 1)
  - You compare your final chi-square statistic against a standard 'critical values' table using you df and chosen alpha level (usually a = .05)
  - If your calculated chi-square is greater than the critical value in the table, the difference is statistically significant
- If your contingency table is larger than 2 x 2 and the chi-square is significant, you must run a post how pairwise comparison to find out exactly which groups differ from each other

---

6.3 Non-parametric tests for ordinal data

Statistical tests that make no assumptions about the probability distribution of the population. They are more flexible than parametric tests, but if you need precise interval or ratio data then they are not precise enough because they collapse the differences into simple ranks.

- Typically used to analyze ordinal data (and sometimes interval data)
- Choosing the right test depends on two things: (1) How many test conditions you have, and (2) Whether your design is within-subjects (correlated samples) or between-subjects (independent samples)

Data Setup	Parametric test	Non-Parametric test
1 Variable 2 Categories Between Subjects	independent t-test	Mann-Whitney U test
1 Variable 2 Categories Within-Subjects	paired t-test	Wilcoxon Signed Rank Test
1 Variable >2 Categories Between Subjects	One-way ANOVA	Kruskal Wallis Test
1 Variable >2 Categories Within Subjects	repeated measures ANOVA	Friedman test Mood's median test
1 Variable (Correlation)	Pearson's r	Spearman's ρ (rho)

We get this on a cheat sheet

1. *Example 1: Mann-Whitney U test*
    - Ex. Comparing the political leaning (rated 1 to 10) of Mac users vs. PC users
        - Since different people are in each group, it is a between-subjects design
2. *Example 2: Wilcoxon Signed-Rank test*
    - Ex. Comparing the 'cool appeal' (rated 1 to 10) of Media Player A vs. Media Player B
        - Because the exact same users tested both players, it is a within-subjects design
3. *Example 3: Kruskal-Wallis test*
    - Ex. Testing how acceptable a new GPS design is across three distinct age groups (20s, 30s and 40s)
        - It calculates an H statistic (which follows a chi-square distribution)
        - Because there are 3+ conditions, you must run a post hoc pairwise comparisons test to find out which groups differ
4. *Example 4: Friedmann test*
    - Ex. Having the same group of users rate the 'quality of results' (1 to 100) across four different search engine interfaces (A, B, C, and D)
        - It also calculates an H statistic and you have to tun a post hoc pairwise comparison if it's statistically significant
5. *Discussion*
    - There is a 'gray area' in HCI statistics regarding questionnaire data. If you consider Likert scales to be linear (Interval data), reporting the Mean (average) is perfectly valid. If you believe human responses are strictly non-linear and subjective (Ordinal data), it is technically more accurate to report the Median (middle value) or Mode (most common value)

6.4 Parametric versus non-parametric tests

Choosing between statistical tests often comes down to balancing statistical power (parametric) against measurement consistency (non-parametric)

- Parametric tests assume your ratio data comes from a perfectly 'normal distribution' (a perfect bell curve). Real-world human performance data is never perfectly normal; it's always skewed or messy, you have three choices when the data isn't 'normal':
    1. Proceed with the parametric test anyway: many researchers do this because these tests are highly robust, which means that they can handle violations of normality and still produce accurate results without sacrificing power
    2. Transform/Clean the data: you can use mathematical functions (like a log function) to force the data to look 'normal' before running the parametric test (this can distort the actual experimental effect)
    3. Switch to a non-parametric test: after realizing that their ratio data is not normal they switch to a non-parametric test to be safe
        - by doing so the test throws away the exact numbers and turns them into simple ranks, you lose massive amounts and statistical power.

    When in doubt with ratio data, run both tests. If they point to the exact same conclusion (which they usually do), report the parametric ANOVA because it retains all your data's precision.

Modeling Interaction (Chapter 7)

A model is a simplification of reality, it reduces complex phenomena (like human behavior or physical spaces) in to a simpler form so researchers can explore, test, and manipulate them without having to build the real thing. There are different kinds of models:

- Descriptive models: On one end, these use loose verbal analogy, metaphor, or visual representation to describe a phenomenon (ex. an architect's physical scale model of a building)
- Predictive models: On the other end, these use formal, closed-form mathematical equations to predict a phenomenon (ex. the amount it takes a user to rapidly move their cursor (or finger) to select a target on a screen (like clicking a button) - Fitts' Law)

7.1 Descriptive models

They let designers think critically about a problem, expose weaknesses or biases in current designs, and weigh the advantages of new ideas before building them.

1. *Delineating a problem space*

   The most basic rule of descriptive model is to take a vast, uncharted topic and break it down into structured categories

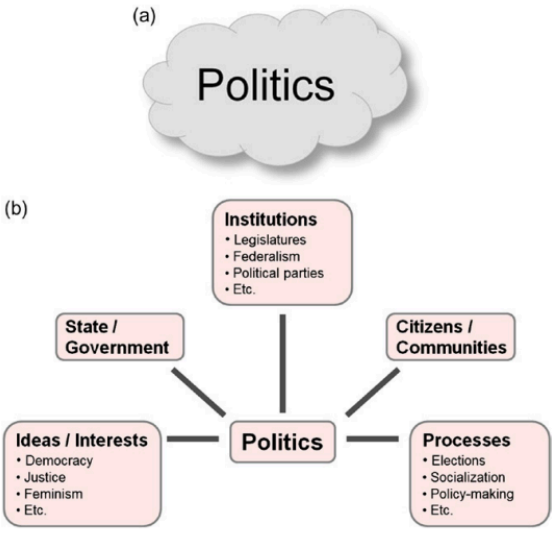


FIGURE 7.1

Politics: (a) The "big fuzzy cloud" model. (b) A descriptive model delineating the problem space.

By breaking "Politics" down into "Institutions," "Processes," etc., we are empowered to study specific relationships. HCI researchers do this exact same thing to map out digital interfaces.

2. *Quadrant model of groupware*

   How do we understand Computer Supported Collaborative Work (CSCW)?

	Same Time	Different Times
Same Place	Copy boards PC Projectors Facilitation Services Group Decision Room Polling Systems	Shared Files Shift Work Kiosks Team Rooms Group Displays
Different Places	Conference Calls Graphics and Audio Screen Sharing Video Teleconferencing Spontaneous Meetings	Group Writing Computer Conferencing Conversational Structuring Forms Management Group Voice Mail

FIGURE 7.2

Quadrant model of groupware.

This model visually divided software using a 2 x 2 matrix based on Space and Time. By plotting tasks here, developers can easily identify what specific features a new piece of software needs to succeed in that quadrant.

3. *Key-action model (KAM)*

- How are keys on a standard keyboard organized and used?

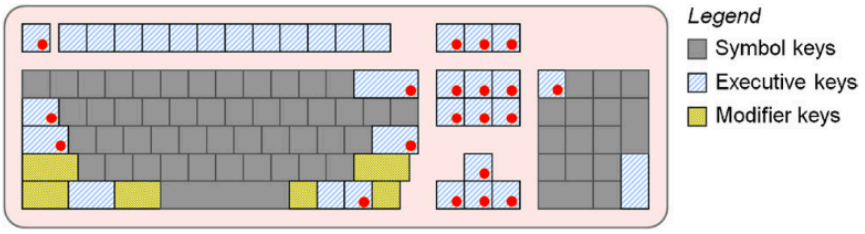


FIGURE 7.3

The key-action model (KAM) illustrated. Keys marked with (red) dots are executive keys that are not mirrored on both sides.

The descriptive model categorized keys into three types: Symbol, Executive (Enter, Esc), and Modifier (Shift). When looking at this visual map researchers saw that it had a massive right-hand bias.

4. *Model of bimanual control*
   - How do humans naturally divide tasks between their two hands?

Hand	Role and Action
Non-preferred	• Leads the preferred hand • Sets the spatial frame of reference for the preferred hand • Performs coarse movements
Preferred	• Follows the non-preferred hand • Works within established frame of reference set by the non-preferred hand • Performs fine movements

FIGURE 7.4

Guiard's model of bimanual control.

(From Guiard, 1987)

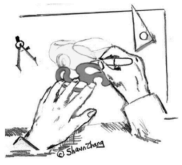


FIGURE 7.5

Two-handed interaction paradigm.

(Sketch courtesy of Shawn Zhang)

This model outlines that the non-preferred hand leads and sets the frame of reference, while the preferred hand follows and does precise work.

Task	Characteristics
Scrolling	• Precedes/overlaps other tasks • Sets the frame of reference • Minimal precision needed (coarse)
Selecting, editing, reading, drawing, etc.	• Follows/overlaps scrolling • Works within frame of reference set by scrolling • Demands precision (fine)

FIGURE 7.6

Relationship between scrolling and common GUI tasks.

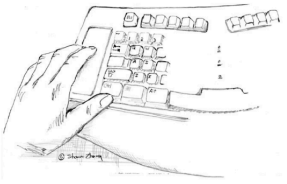


FIGURE 7.7

Scrolling interface example.

(Sketch courtesy of Shawn Zhang)

When mapped to computer tasks, this model shows that scrolling (setting the frame of reference) should naturally belong to the non-preferred hand.

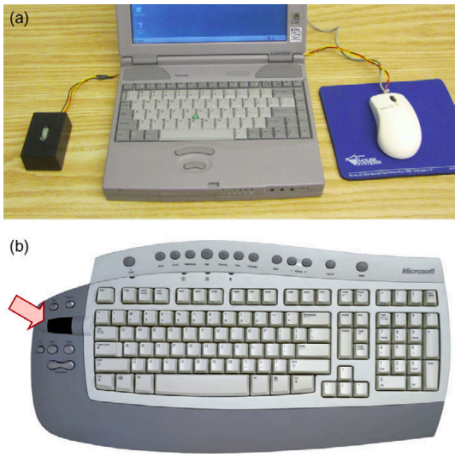


FIGURE 7.8

(a) The author's re-engineered Microsoft Intellimouse with the wheel on left side of the keyboard. (b) Microsoft Office Keyboard with scrolling wheel on the left side.

Because the descriptive model proved that standard right-handed 'wheel mice' violate natural human bimanual control, it inspired this design on the left side.

5. *Three-state model for graphical input*
    - How do pointing devices (mice, touchpads) actually navigate system states?

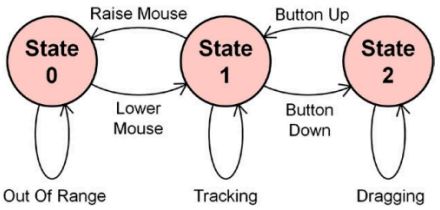


FIGURE 7.9

Buxton's three-state model for graphical input.

(Adapted from Buxton, 1990)

This model describes pointing devices through three simple state transitions: State 0 (Out of range/Clutching - lifting the mouse up), State 1 (Tracking - just moving the mouse around), and State 2 (Dragging - click and hold something).

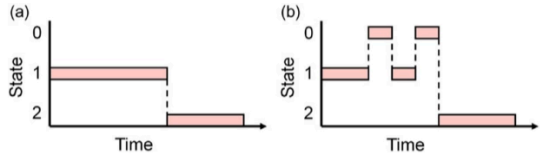


FIGURE 7.10

State transitions for dragging tasks: (a) Mouse. (b) Lift-and-tap touchpad. Dragging begins upon entering State 2.

(Adapted from MacKenzie and Oniszczak, 1998)

Researchers used Buxton's model to compare a standard mouse to early 'lift-and-tap' touchpads, it forces the user into messy, extra state transitions (1 - 0 - 1 - 2) that easily get confused with simple clutching. This led to touchpads that let users 'press harder' to click.

7.2 Predictive Models

Predictive models are entirely quantitative. To predict an outcome, you need a predictor variable (the independent variable). In these equations the predictor must be a continuous, ratio-scale number (ex. distance to target, system lag, etc.), you can't use nominal data (like mouse vs. keyboard) in a mathematical equation.

1. *Linear regression model*
    - The statistical process to build the prediction equation (y = mx + b). It takes a scatter of actual human data points and finds the 'best-fitting straight line' that minimizes the distance between the points and the line.

(a)

Participant	Stylus Tapping Speed (wpm)	Touch Typing Speed (wpm)
P1	18.2	42
P2	23.6	44
P3	26.0	32
P4	20.3	50
P5	20.3	36
P6	17.1	33
P7	24.0	74
P8	14.7	22
P9	20.3	31
P10	19.7	33
P11	22.4	25
P12	13.1	19

(b)

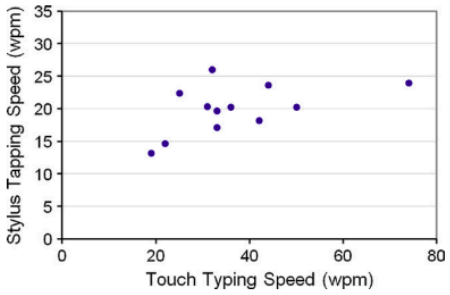


FIGURE 7.12

Relationship between stylus-tapping speed and touch-typing speed: (a) Data. (b) Scatter plot.
(Adapted from MacKenzie and Zhang, 2001, Fig. 4)

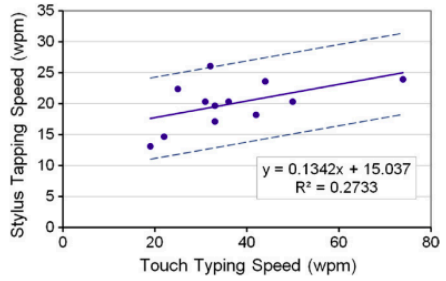


FIGURE 7.13

Scatter plot from Figure 7.12b embellished with linear regression line, prediction equation, squared coefficient of correlation (R^2) and dashed lines showing the 95% confidence interval.

These plots visually map the relationship between two variables. The line drawn through the dots is the predictive model. The R^2 value (squared correlation coefficient) tells you how good the model is by explaining the percentage of variance.

2. *Fitts' Law*
    - Predicts the time required to rapidly move a cursor or finger to a target. Fitts' Law proves that task difficulty relies entirely on two things: the distance to the target (A) and the width of the target (W).
        - It implies that closer, larger targets are faster and easier to select, while distant, smaller targets are slower and more error-prone

A (pixels)	*W* (pixels)	*ID* (bits)	Mouse				RemotePoint			
			W_e (pixels)	ID_e (bits)	*MT* (ms)	*TP* (bits/s)	W_e (pixels)	ID_e (bits)	*MT* (ms)	*TP* (bits/s)
40	10	2.32	11.23	2.19	665	3.29	13.59	1.98	1587	1.25
40	20	1.58	19.46	1.61	501	3.21	21.66	1.51	1293	1.17
40	40	1.00	40.20	1.00	361	2.76	37.92	1.04	1001	1.04
80	10	3.17	10.28	3.13	762	4.11	10.08	3.16	1874	1.69
80	20	2.32	18.72	2.40	604	3.97	25.21	2.06	1442	1.43
80	40	1.58	35.67	1.70	481	3.53	37.75	1.64	1175	1.40
160	10	4.09	10.71	3.99	979	4.08	10.33	4.04	2353	1.72
160	20	3.17	21.04	3.11	823	3.77	19.09	3.23	1788	1.81
160	40	2.32	41.96	2.27	615	3.69	35.97	2.45	1480	1.65
		Mean	23.25	2.38	644	3.60	23.51	2.35	1555	1.46

FIGURE 7.16

Experiment conditions (*A*, *W*, *ID*) and participants' responses (W_e, ID_e, *MT*, *TP*) for the standard mouse and the RemotePoint.

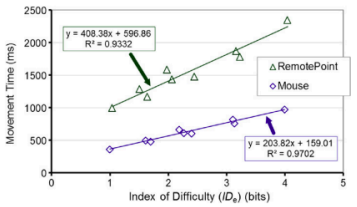


FIGURE 7.17

Scatter plot and regression lines for data in Figure 7.16b.

(Adapted from MacKenzie and Jusoh, 2001)

When comparing two devices, a steeper slope on the regression line means the device is slower/worse.

3. *Hick-Hyman Law (choice reaction time)*
   - Predicts how long it will take a user to react to a stimulus and choose the correct response out of n possible alternatives
     - If certain choices appear more frequently than others, the user's reaction time actually drops because there is less overall uncertainty in the system
4. *The Keystroke-Level Model (KLM)*
   - An engineering model that predicts the exact, error-free execution time for an expert user to complete a specific task on an interface

Operator	Description	Associated Time
K	Keystroke, typing one letter, number, etc. or function key like 'CTRL', 'SHIFT'	Expert typist (90 wpm): 0.12 sec Average skilled typist (55 wpm): 0.20 sec Average non-secretarial typist (40 wpm): 0 .28 sec Worst typist (unfamiliar with keyboard): 1.2 sec
H	'Homing', moving the hand between mouse and keyboard	0.4 sec
B / BB	Pressing / clicking a mouse button	0.1 sec / 2*0.1 sec
P	Pointing with the mouse to a target	0.8 to 1.5 sec with an average of 1.1 sec Can also use Fitts' Law
D(n$_D$, l$_D$)	Drawing n_D straight line segments of length l_D	$0.9*n_D + 0.16*l_D$
M	Subsumed time for mental acts; sometimes used as 'look-at'	1.35 sec (1.2 sec according to [Olson and Olson 1995])
R(t) or ***W(t)***	System response (or 'work') time, time during which the user cannot act	Dependent on the system, to be determined on a system-by-system basis

We get this on a cheat sheet. You always take the average amount of seconds when there are multiple choices (and the average non-secretarial typist).

   - You break the task down into a sequence of primitive operators (from the photo) and sum up their individual times
5. *Skill acquisition*
   - Predicts how human performance (time or speed) improves with practice
   - The relationship between skill and practice is non-linear. In the beginning, a small amount of practice yields massive improvements, but as the user becomes an expert, those returns diminish, and the same amount of practice only yields tiny improvements.
6. *Multiple regression*
   - Builds a predictive model using more than one predictor variable (ex. predicting how fast someone learns a game based on both Age and Daily Computer Use)
   - When you add more predictor variables to an equation, the correlation will always increase. A higher correlation does not automatically mean it is a 'better' model, the variables you add must actually offer logical, explanatory power

Writing and Publishing a Research Paper (Chapter 8)

8.1 Conference vs Journal papers

Aspect	Conference Paper	Journal Paper

Length	Short (6-10) pages, strict format	Longer, flexible format
Review	Fast (months), deadline-driven	Slow (many months-year), multiple rounds of revision
Focus	Novel developments, preliminary findings	In depth analysis, mature results
Presentation	Oral talk at conference	Read only
Prestige in HCI	Top venues like ACM SIGCHI rival journals	Traditional gold standard

8.2 Parts of a research paper

Title

- Short, identifies subject area, narrows scope. Often catchy. Two-part titles with colon common.

Abstract

- Single paragraph, ~150 words, written last.
- Two objectives: tell reader (1) what was done, (2) what was found (with specific results — not just "see paper").
- No citations, no introductory material (that belongs in the Introduction).
- Most important section — decides if paper gets read or accepted.

Keywords

- For database indexing and searching.

Introduction

- Frames the problem, motivates the work, states the research question, previews contribution.

Related Work

- Literature review — cites prior work, positions your contribution relative to existing research.

Method

- How the study was done. Includes: participants, apparatus, design (factors, levels, within/between), procedure, tasks.
- Should be detailed enough to allow replication.

Results and Discussion

- Results: statistical outcomes, tables, charts, significance tests.
- Discussion: interpret, relate to hypotheses, discuss limitations.

Conclusion

- Summarizes contribution, restates key findings, suggests future work. One paragraph, like an abstract but contribution-focused.

References

- Only papers cited in text; all cited papers must appear here; format per venue style.

8.3 Preparing the manuscript

- Follow the venue's formatting guidelines (especially for conferences — strict).
- Use LaTeX or Word with the provided template.
- Use clear figures and tables — captions must stand alone.
- Edit heavily for clarity and conciseness. Spell-check.
- Use consistent terminology